Progress Report I:

A* Agent / Search Algorithm Implementation

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1. Objective

This report details the successful implementation of the second major component of the DTRA, as outlined in our project proposal.

The primary objective for this milestone was to fulfill

Progress Report I:

Agent/search algorithm implementation with working example.

This involved building the "Decider" brain of our agent. This agent's job is to:

1. Receive a **"Danger Score"** (a probability, e.g., 0.95) from the "Detector" (built in Progress Report II).
2. Model the response as a **cost-minimization problem** (as specified in Section 7 of our proposal).
3. Use an **A* search** logic to find the "cheapest" (best) action from the list: Ignore, Log, Block, or Isolate.
4. Provide a **"working example"** that proves the agent's logic is sound.

2. Methodology & Implementation (The A* Agent)

As defined in our proposal, we modeled our response selection as a "cost-minimization problem." We implemented this using the A* search formula:

$f(n) = g(n) + h(n)$

Where:

- $f(n)$ = **Total Cost** (We want to find the action with the *minimum* $f(n)$).
- $g(n)$ = **Business Cost** (The "disruption" cost we *assign*).
- $h(n)$ = **Security Risk** (The "danger" cost we *calculate*).

A. g(n): "Business Cost" Implementation

We first defined a set of "price tags" (our BUSINESS_COSTS rules) to teach the agent our business priorities. These are our "invented" costs for how disruptive each action is:

- Ignore: 0 (Free)
- Log: 2 (Very Cheap)
- Block: 20 (Medium Disruption)
- Isolate: 80 (High Disruption)

B. h(n): "Security Risk" Implementation

We then implemented a function to *calculate* the security risk based on the "Danger Score" (from Report II). The logic is as follows:

- **Risk of Ignore:** The risk is the *full* "Danger Score." Ignoring a 95% threat has a risk of 95.
- **Risk of Block:** The risk is the *uncertainty* (the chance of a false positive). Blocking a 95% threat has a risk of only (1.0 - 0.95) * 20 = 1.0.
- **Risk of Isolate:** The risk is 0 (the threat is 100% contained).

3. "Working Example" (The Output)

We ran the implemented agent against three distinct test scenarios: a "Leaf" (false positive), a "Robber" (true threat), and an "Ambiguous" threat.

The following output is our "working example."

```
--- Running A* 'Decider' Agent: Working Examples ---
--- 'Decider' is thinking about a 1.0% Danger Score ---
  - Action: Ignore   | Total Cost:   1.00 (g_cost=0  + h_cost=1.00  )
  - Action: Log      | Total Cost:   2.90 (g_cost=2  + h_cost=0.90  )
  - Action: Block    | Total Cost:  39.80 (g_cost=20 + h_cost=19.80 )
  - Action: Isolate  | Total Cost:  80.00 (g_cost=80 + h_cost=0.00  )
  ==> Best Action for 'Leaf': Ignore (Total Cost: 1.00)

--- 'Decider' is thinking about a 95.0% Danger Score ---
  - Action: Ignore   | Total Cost:  95.00 (g_cost=0  + h_cost=95.00 )
  - Action: Log      | Total Cost:  87.50 (g_cost=2  + h_cost=85.50 )
  - Action: Block    | Total Cost:  21.00 (g_cost=20 + h_cost=1.00  )
  - Action: Isolate  | Total Cost:  80.00 (g_cost=80 + h_cost=0.00  )
  ==> Best Action for 'Robber': Block (Total Cost: 21.00)

--- 'Decider' is thinking about a 60.0% Danger Score ---
  - Action: Ignore   | Total Cost:  60.00 (g_cost=0  + h_cost=60.00 )
  - Action: Log      | Total Cost:  56.00 (g_cost=2  + h_cost=54.00 )
  - Action: Block    | Total Cost:  28.00 (g_cost=20 + h_cost=8.00  )
  - Action: Isolate  | Total Cost:  80.00 (g_cost=80 + h_cost=0.00  )
  ==> Best Action for 'Ambiguous': Block (Total Cost: 28.00)
```

4. Results Analysis

This output clearly demonstrates that our "Decider" agent is working logically and fulfilling its objectives.

- **Test 1 (1.0% "Leaf"):** The agent's "thinking" shows that Ignore has the lowest "Total Cost" (1.00) by a huge margin. The agent correctly chose to **Ignore** the alert. **This is the core "Alert Fatigue" solution** from our proposal.
- **Test 2 (95.0% "Robber"):** The agent saw that Ignore was now the *most expensive* action (Total Cost: 95.00). It correctly calculated that **Block** (Total Cost: 21.00) was the "cheapest" and most logical action, balancing the medium "Business Cost" (20) with the tiny "Security Risk" (1.0). This is our "autonomous response" in action.
- **Test 3 (60.0% "Ambiguous"):** The agent again showed it is "safety-first." The cost to Ignore (60.00) was more than *twice* the cost to Block (28.00). The agent correctly chose to Block the "ambiguous" threat, deciding it was "cheaper" to be safe than sorry.

5. Conclusion

This milestone is a complete success. We have:

1. Successfully **implemented the A* search algorithm** as our "Decider" agent, fulfilling Section 7 of our proposal.
2. Programmed the agent to take four distinct actions (Ignore, Log, Block, Isolate), fulfilling Section 3 of our proposal.
3. Provided a clear **"working example"** with three test cases that prove the agent's logic is sound and effective.